



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

December 2010
Assessment Task 1
Year 11

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question if full marks are to be awarded.
- All answers to be given in simplified exact form unless otherwise stated.
- Marks may not be awarded for messy or badly arranged work

Total Marks – 74

- Attempt questions 1-4
- All questions are **NOT** of equal value.
- Start each new question in a separate answer booklet.
- Hand in your answers in 4 separate bundles:

Question 1, Question 2, Question 3 and Question 4

Examiner: *R Boros*

Start a new booklet.**Question 1 (18 marks).****Marks**

- a) Express $\frac{7\pi}{18}$ in degrees. 1
- b) Find the value of k if $(x-2)$ is a factor of $P(x) = x^4 - 3x^3 + kx^2 - 4$ 1
- c) Solve for x , $\frac{x+1}{x-3} \geq 2$ 2
- d) Find the acute angle, to the nearest degree, between the lines $y = 2x + 3$ and $x + y = 0$ 2
- e) Find the general solution of $2\cos\theta - 1 = 0$, where θ is in radians. 3
- f) Prove that $\tan\left(\frac{\pi}{4} + A\right) + \tan\left(\frac{\pi}{4} - A\right) = 2\sec 2A$. 3
- g) (i) In how many ways can 3 men and 3 women be arranged in a circle? 1
- (ii) How many of these arrangements have at least 2 women sitting next to each other? 2
- h) Find the co-ordinates of the point P which divides the interval joining $A(-1,-3)$ and $B(3,7)$ externally in the ratio 5:3. 3

End of Question 1

Start a new booklet.**Question 2 (18 Marks).****Marks**

- a) The roots of a cubic polynomial equation are 0, 1 and 3; and the coefficient of x^3 is 2. Find the polynomial in full expanded form. 3
- b) Solve for n : $2 \times^n C_4 = 5 \times^n C_2$. 2
- c) If α, β and γ are the roots of $x^3 - 3x + 1 = 0$, find;
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta\gamma$ 1
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- d) (i) Express $\sqrt{3} \cos x - \sin x$ in the form $R \cos(x + \alpha)$, $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- (ii) Hence, find the general solution for $\sqrt{3} \cos x - \sin x = 1$. 2
- e) α and β are acute angles, such that $\cos \alpha = \frac{3}{5}$ and $\sin \beta = \frac{1}{\sqrt{5}}$. Without finding the size of either angle, show that $\alpha = 2\beta$. 2
- f) Using the identity $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$, solve the equation $\sin 3\theta = 2\sin \theta$, where $0 \leq \theta \leq 2\pi$. 3

End of Question 2

Start a new booklet.**Question 3 (18 marks).****Marks**

- a)** (i) Show that the equation $\sin x^\circ - 3\cos x^\circ = \tan\left(\frac{x}{2}\right)^\circ$ may be written as 3

$$t^3 - 3t^2 - t + 3 = 0, \text{ where } t = \tan\left(\frac{x}{2}\right)^\circ$$

- (ii) Hence, find all 3 solutions of the equation $\sin x^\circ - 3\cos x^\circ = \tan\left(\frac{x}{2}\right)^\circ$, for 3

$0^\circ \leq x \leq 360^\circ$. Give answers to the nearest minute, if necessary.

- b)** Six identical yellow discs and 4 identical blue discs are placed in a straight line. 1
How many arrangements are possible?

- c)** (i) In how many ways can a committee of 2 Australians, 2 New Zealanders 2
and 1 Nauruan be chosen from 6 Australians, 7 New Zealanders and 3 Nauruans?

- (ii) In how many of these ways do 2 friends, a particular Australian and a 1
particular New Zealander, belong to the committee?

- d)** For the parabola $x^2 = 12y$: 2
(i) Derive the equation of the tangent at $(6t, 3t^2)$.

- (ii) Find the equations of the two tangents that pass through the point $(5, -2)$. 3

- e)** Find the value of the constants p and q if $x^2 - 4x + 3$ is a factor of 3
 $x^3 + px^2 - x + q$.

End of Question 3

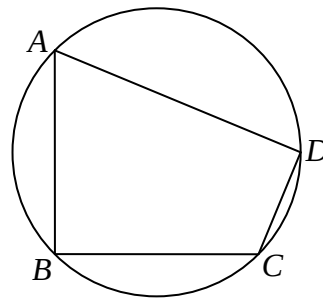
Start a new booklet.**Question 4 (20 marks).****Marks**

- a) In how many ways can 8 different books be arranged in a row such that 3 particular books are always together?

2

- b) $ABCD$ is a cyclic quadrilateral. One of its properties is that opposite angles are supplementary. Show that $\tan A + \tan B + \tan C + \tan D = 0$

2



- c) Eight people including James and Sarah are to be seated around a table. How many arrangements are possible if James and Sarah do not wish to sit next to each other.

2

- d) The straight line $y = mx + b$ meets the parabola $x^2 = 4ay$ at the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$

- (i) Find the equation of the chord PQ and, hence or otherwise, show that

2

$$pq = -\frac{b}{a}$$

- (ii) Prove that $p^2 + q^2 = 4m^2 + \frac{2b}{a}$

2

- (iii) Given that the equation of the normal to the parabola at P is

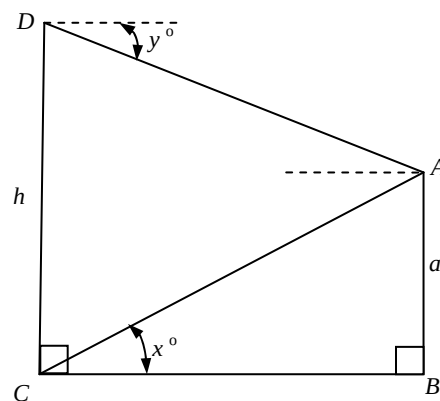
2

$x + py = 2ap + ap^3$, and that, N , the point of intersection of the normals at P and Q has the co-ordinates $[-apq(p+q), a(2 + p^2 + pq + q^2)]$, express these co-ordinates in terms of a , m and b .

- (iv) Suppose that the chord PQ is free to move while maintaining a fixed gradient. Find the locus of N , and show that this locus is a straight line. Verify that this line is a normal to the parabola.

2

Question 4 continues on next page

Question 4 (continued).**e)**

From the foot of a tower CD , the angle of elevation of a building AB , ' a ' metres high, is x° . From the top of the tower, D , the angle of depression to the top of the building, A is y° . Show that the height, ' h ', of the tower is given by:

6

$$h = \frac{a \sin(x + y)}{\sin x \cos y}$$

End of Question 4**End of Examination**

$$1) a) \frac{7\pi}{18} \times \frac{180}{\pi} = \underline{70^\circ}$$

$$(b) P(x) = x^4 - 3x^3 - kx^2 - 4$$

$$P(2) = (2)^4 - 3(2)^3 - k(2)^2 - 4 = 0$$

$$4k - 12 = 0$$

$$\underline{k = 3}$$

$$(c) \frac{x+1}{x-3} \geq 2 \quad x \neq 3.$$

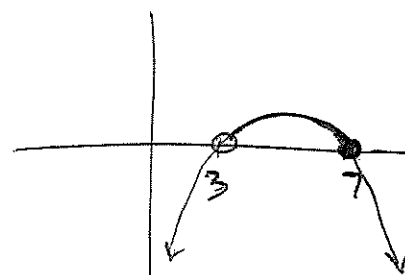
$$(x+1)(x-3) \geq 2(x-3)^2$$

$$(x+1)(x-3) - 2(x-3)^2 \geq 0$$

$$(x-3)[x+1 - 2(x-3)] \geq 0$$

$$(x-3)(-x+7) \geq 0$$

$$\underline{3 < x \leq 7}$$



$$(d) y = 2x + 3 \quad x + y = 0$$

$$m_1 = 2 \quad y = -x$$

$$m_2 = -1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{(2) - (-1)}{1 + (2)(-1)} \right|$$

$$= 3$$

$$\underline{\theta = 72^\circ} \text{ (to nearest degree)}$$

$$(e) \quad 2 \cos \theta - 1 = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\frac{S}{T} \mid \begin{matrix} A \checkmark \\ c \checkmark \end{matrix}$$

$$\cos \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{3}$$

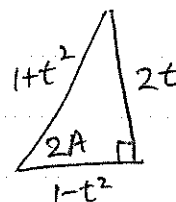
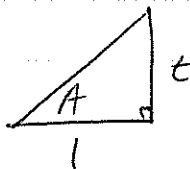
$$\theta = 2\pi n \pm \frac{\pi}{3} \quad \text{where } n \text{ is an integer.}$$

$$(f) \quad \text{LHS} = \tan\left(\frac{\pi}{4} + A\right) + \tan\left(\frac{\pi}{4} - A\right)$$

$$= \frac{\tan \frac{\pi}{4} + \tan A}{1 - \tan \frac{\pi}{4} \tan A} + \frac{\tan \frac{\pi}{4} - \tan A}{1 + \tan \frac{\pi}{4} \tan A}$$

$$\tan \frac{\pi}{4} = 1$$

$$\text{let } \tan A = t$$



$$= \frac{1+t}{1-t} + \frac{1-t}{1+t}$$

$$= \frac{(1+t)^2 + (1-t)^2}{1-t^2}$$

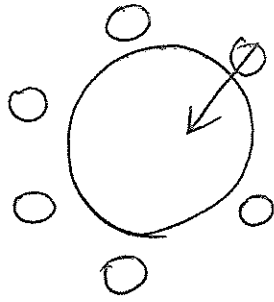
$$= \frac{1 + \cancel{2t} + t^2 + 1 - \cancel{2t} + t^2}{1-t^2}$$

$$= \frac{2(1+t^2)}{1-t^2}$$

$$= 2 \sec 2A$$

$$= \text{RHS.}$$

(g) i)



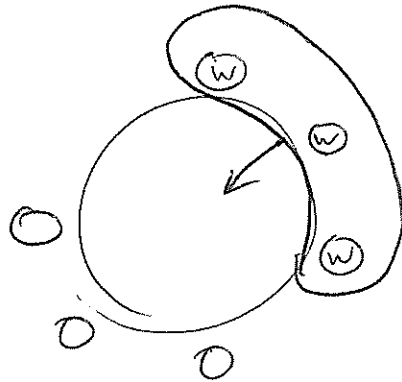
$$1 \times 5! = \underline{120}$$

(ii) Separated $1 \times 3! \times 2! = 12$

At least 2: $120 - 12 = \underline{108}$

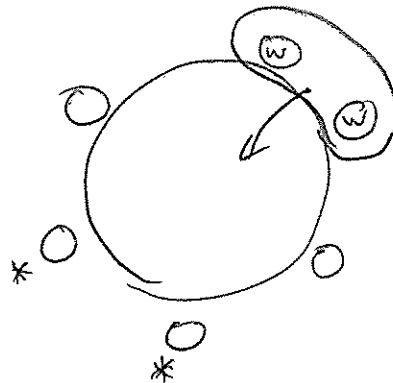
OR

3 women together:



$$3! \times 1 \times 3! = 36$$

2 women together:



$$\frac{3 \times 2 \times 1 \times 2 \times 3!}{3P_2} = 72$$

* possibilities for other women.

At least 2: $36 + 72 = \underline{108}$

(h) $A(-1, -3) \longleftrightarrow B(3, 7)$
 $5: -3$

$$P\left(\frac{-3(-1) + 5(3)}{5 + (-3)}, \frac{-3(-3) + 5(7)}{5 + (-3)}\right)$$

$$P(9, 22).$$

EXTENSION 1 SOLN'S

QUESTION 2.

$$\begin{aligned} \text{a) } P(x) &= 2x(x-1)(x-3) \\ &= (2x^2 - 2x)(x-3) \\ &= 2x^3 - 8x^2 + 6x \end{aligned}$$

OR.

$$\alpha + \beta + \gamma = 0 + 1 + 3 = 4$$

$$4 = -\frac{b}{a}$$

$$\text{But } a = 2$$

$$\therefore \boxed{b = -8}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 0 + 0 + 3 = 3$$

$$\therefore 3 = \frac{c}{a}$$

$$3 = \frac{c}{2}$$

$$\boxed{c = 6}$$

$$\alpha\beta\gamma = 0 \times 1 \times 3 = 0$$

$$\therefore 0 = -\frac{d}{a}$$

$$\therefore \boxed{d = 0}$$

$$\text{Given } a = 2$$

$$\therefore P(x) = 2x^3 - 8x^2 + 6x.$$

$$\text{b) } 2 \times {}^nC_4 = 5 \times {}^nC_2.$$

$$2 \times \frac{n!}{4!(n-4)!} = 5 \times \frac{n!}{2!(n-2)!}$$

$$2 \cdot 2!(n-2)! = 5 \cdot 4!(n-4)!$$

$$\frac{(n-2)!}{(n-4)!} = \frac{5 \cdot 4!}{2 \cdot 2!}$$

$$(n-3)(n-2) = \frac{5 \times 4 \times 3 \times 2}{2 \times 2}$$

$$n^2 - 5n + 6 = 30$$

$$n^2 - 5n - 24 = 0$$

$$(n-8)(n+3) = 0$$

$$n = 8, -3 \quad n > 0$$

$$\therefore \boxed{n = 8}$$

$$\begin{aligned} \text{c) (i) } \alpha + \beta + \gamma &= -\frac{b}{a} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{(ii) } \alpha\beta\gamma &= -\frac{d}{a} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{(iii) } \frac{\alpha\gamma + \beta\gamma + \alpha\beta}{\alpha\beta\gamma} &= \frac{-3}{-1} \\ &= 3. \end{aligned}$$

$$\text{d) (i) } \sqrt{3} \cos x - \sin x = R \cos(x + \alpha)$$

$$\begin{aligned} R &= \sqrt{3+1} \\ &= 2 \end{aligned}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore R \cos(x + \alpha) = 2 \cos\left(x + \frac{\pi}{6}\right)$$

$$d) (ii) \quad 2\cos\left(x + \frac{\pi}{6}\right) = 1$$

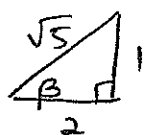
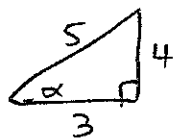
$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$$

$$x = 2n\pi \pm \frac{\pi}{3} - \frac{\pi}{6}$$

$$\text{OR } x = 2n\pi + \frac{\pi}{6}, 2n\pi - \frac{\pi}{2}$$

e)



$$\cos \alpha = \frac{3}{5} \quad \sin \beta = \frac{1}{\sqrt{5}}$$

$$\sin 2\beta = 2\sin\beta\cos\beta$$

$$= 2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}}$$

$$= \frac{4}{5}$$

$$\sin \alpha = \frac{4}{5}$$

$$\sin \alpha = \sin 2\beta$$

$$\therefore \alpha = 2\beta.$$

$$f) \quad \sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\text{But } \sin 3\theta = 2\sin\theta$$

$$\therefore 3\sin\theta - 4\sin^3\theta = 2\sin\theta$$

$$\therefore \sin\theta - 4\sin^3\theta = 0$$

$$\sin\theta(1 - 4\sin^2\theta) = 0$$

$$\sin\theta = 0, \quad 1 - 4\sin^2\theta = 0$$

$$\therefore \theta = 0, \pi, 2\pi$$

$$\sin^2\theta = \frac{1}{4}$$

$$\sin\theta = \pm \frac{1}{2}$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\sin\theta = -\frac{1}{2}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

Marking Scale Comments

2 a) 2 marks for $2x^3 - 4x^2 + 3x$
2 marks for not fully expanding.
1 mark for each coefficient.

b) 1 mark for $\frac{2 \times n!}{4!(n-4)!} = 5 \times \frac{n!}{2!(n-2)!}$

$1\frac{1}{2}$ marks if they almost got the answer of $n=8$
because only made a ~~small~~ small error

c) (i) } no half marks, 1 mark only.
(ii)

(ii) $\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{2\beta\gamma}$ + wrong answer - 1 mark only.

If carried on with previous wrong answer, should not have been penalized again - 2 marks.

d) (i) $\alpha = 30^\circ$ instead of $\alpha = \frac{\pi}{6}$ - lost $\frac{1}{2}$ mark.
 $\alpha = -\frac{\pi}{6}$ - $\frac{1}{2}$ mark.

(ii) 2 marks - $2n\pi \pm \frac{\pi}{3} - \frac{\pi}{6}$
- $2n\pi + \frac{\pi}{6}, 2n\pi - \frac{\pi}{6}$

1 mark - $2n\pi \pm \frac{\pi}{6}$

1 mark - $(x + \frac{\pi}{6}) = 2n\pi \pm \frac{\pi}{3}$

$\frac{1}{2}$ mark - $2n\pi \pm \alpha$

2010 Extension Mathematics Task 1:
Solutions— Question 3

3. (a) (i) Show that the equation $\sin x^\circ - 3 \cos x^\circ = \tan \left(\frac{x}{2} \right)^\circ$ may be written as $t^3 - 3t^2 - t + 3 = 0$, where $t = \tan \left(\frac{x}{2} \right)^\circ$. 3

Solution:

$$\frac{2t}{1+t^2} - \frac{3(1-t^2)}{1+t^2} = t,$$

$$2t - 3 + 3t^2 = t + t^3,$$

i.e. $t^3 - 3t^2 - t + 3 = 0$.

- (ii) Hence find all three solutions of the equation $\sin x^\circ - 3 \cos x^\circ = \tan \left(\frac{x}{2} \right)^\circ$, for $0^\circ \leq x^\circ \leq 360^\circ$. Give answers to the nearest minute if necessary. 3

Solution: Method 1

$$t^2(t-3) - (t-3) = 0,$$

$$(t^2-1)(t-3) = 0,$$

$$(t+1)(t-1)(t-3) = 0,$$

so $t = 3, \pm 1$.

$$\frac{x^\circ}{2} \approx 71^\circ 34', 45^\circ, 135^\circ,$$

$$x^\circ \approx 143^\circ 8', 90^\circ, 270^\circ.$$

Solution: Method 2

Put $P(t) = t^3 - 3t^2 - t + 3 = 0$,
 $P(3) = 27 - 3 \times 9 - 3 + 3 = 0$.

$\therefore (t-3)$ is a factor.

$$\begin{array}{r|rrrr} & 1 & -3 & -1 & 3 \\ 3 & & 3 & 0 & -3 \\ \hline & 1 & 0 & -1 & 0 \end{array}$$

$$\therefore P(t) = (t-3)(t^2-1),$$

$$= (t-3)(t-1)(t+1),$$

so $t = 3, \pm 1$.

$$\frac{x^\circ}{2} \approx 71^\circ 34', 45^\circ, 135^\circ,$$

$$x^\circ \approx 143^\circ 8', 90^\circ, 270^\circ.$$

- (b) Six identical yellow discs and four identical blue discs are placed in a straight line. How many arrangements are possible? 1

Solution: $\frac{10!}{6!4!} = 210$.

- (c) (i) In how many ways can a committee of 2 Australians, 2 New Zealanders, and one Nauruan be chosen from 6 Australians, 7 New Zealanders, and 3 Nauruans?

2

Solution: $\binom{6}{2} \binom{7}{2} \binom{3}{1} = 945.$

- (ii) In how many of these ways do two friends, a particular Australian and a particular New Zealander, belong to the committee?

1

Solution: $\binom{5}{1} \binom{6}{1} \binom{3}{1} = 90.$

- (d) For the parabola $x^2 = 12y$:

- (i) Derive the equation of the tangent at $(6t, 3t^2)$.

2

Solution: $\frac{dy}{dx} = \frac{2x}{12},$
 $= \frac{x}{6}.$
 $\therefore$ Tangent is $y - 3t^2 = \frac{6t}{6}(x - 6t),$
 $y - 3t^2 = tx - 6t^2,$
i.e. $y = tx - 3t^2.$

- (ii) Find the equations of the tangents that pass through the point $(5, -2)$.

3

Solution: As they pass through $(5, -2),$
 $-2 = 5t - 3t^2,$
 $3t^2 - 5t - 2 = 0,$
 $(3t + 1)(t - 2) = 0,$
 $\therefore t = 2, -1/3.$
 $\therefore$ When $t = 2,$ tangent is
 $y = 2x - 12,$
or $2x - y - 12 = 0.$
When $t = -1/3,$ tangent is
 $y = \frac{-x}{3} - \frac{1}{3},$
or $x + 3y + 1 = 0.$

(e) Find the value of the constants p and q if $x^2 - 4x + 3$ is a factor of $x^3 + px^2 - x + q$.

3

Solution: Method 1

$$\begin{aligned}
 x^2 - 4x + 3 &= (x - 3)(x - 1). \\
 P(x) &= x^3 + px^2 - x + q, \\
 P(3) &= 27 + 9p - 3 + q = 0, \\
 &\quad 9p + q + 24 = 0 \dots\dots\dots [1] \\
 P(1) &= 1 + p - 1 + q = 0, \\
 &\quad p + q = 0 \dots\dots\dots [2] \\
 [1] - [2]: &\quad 8p + 24 = 0, \\
 &\quad p = -3, \\
 &\quad q = 3.
 \end{aligned}$$

Solution: Method 2

$$\begin{aligned}
 x^3 + px^2 - x + q &= (x - \alpha)(x^2 - 4x + 3), \\
 &= (x - \alpha)(x - 3)(x - 1) \text{ i.e. roots } \alpha, 3, 1. \\
 \Sigma\alpha: &\quad \alpha + 3 + 1 = -p, \\
 &\quad \alpha + 4 = -p. \\
 \Sigma\alpha\beta: &\quad 3\alpha + \alpha + 3 = -1, \\
 &\quad 4\alpha = -4, \\
 &\quad \alpha = -1. \\
 \Sigma\alpha\beta\gamma: &\quad -1 \cdot 3 \cdot 1 = -q, \\
 &\quad q = 3, \\
 &\quad -1 + 4 = -p, \\
 &\quad p = -3.
 \end{aligned}$$

Solution: Method 3

$$\begin{array}{r}
 \begin{array}{r}
 x^2 - 4x + 3 \quad \Bigg) \begin{array}{r}
 x^3 + \quad px^2 \quad - \quad x \quad + \quad q \\
 x^3 - \quad 4x^2 \quad + \quad 3x \\
 \hline
 (p+4)x^2 - \quad 4x \quad + \quad q \\
 (p+4)x^2 - \quad 4(p+4)x + \quad 3(p+4) \\
 \hline
 4px + 12x + \quad q - 3(p+4)
 \end{array}
 \end{array}
 \end{array}$$

$$\begin{aligned}
 \text{Thus } x^3 + px^2 - x + q &= (x^2 - 4x + 3)(x + (p + 4)), \\
 \text{equating coefficients, } 3(p + 4) &= q. \\
 \text{Also, } 4px + 12x + q - 3(p + 4) &= 0, \\
 \text{i.e. } 4(p + 3)x + 3(p + 4) - 3(p + 4) &= 0, \\
 4(p + 3)x &= 0, \\
 p &= -3, \\
 q &= 3(-3 + 4), \\
 &= 3.
 \end{aligned}$$

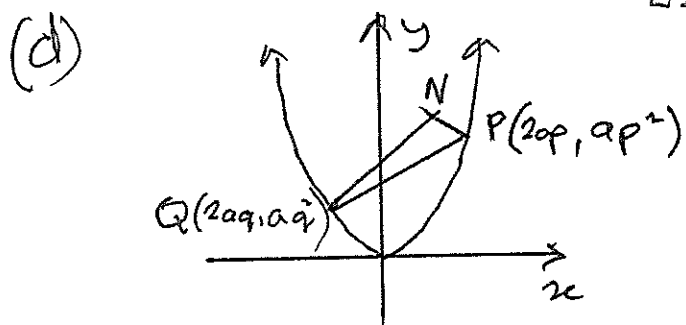
Question 4 (Yr 11 Ext 1)

(a) $6! \times 3! = 4320$ [2]

(b) $\tan A + \tan B + \tan C + \tan D$
 $= \tan A + \tan B + \tan(180^\circ - A)$
 $+ \tan(180^\circ - B)$
 $= \tan A + \tan B - \tan A - \tan B$
 $= 0$ [2]

(c) Total Arrangements = $7!$
 Total with J & S together = $6! \times 2$

$\therefore$ Arrangements w/o J & S together = $7! - 2 \times 6!$
 $= 6!(7 - 2)$
 $= 5 \times 6!$
 $= 3600$ [2]



(i) For PQ: $m = \frac{ap^2 - aq^2}{2ap - 2aq}$
 $= \frac{p+q}{2}$

$\therefore y - ap^2 = \frac{p+q}{2}(x - 2ap)$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$y = \left(\frac{p+q}{2}\right)x - apq \quad \text{--- (1)}$$

or $(p+q)x - 2y - 2apq = 0$

Now (1) may also be written $y = mx + b$

$$\therefore b = -apq$$

$$\text{So } pq = -\frac{b}{a} \quad [2]$$

(ii) $\frac{p+q}{2} = m$

$$\therefore \frac{p^2 + 2pq + q^2}{4} = m^2$$

$$\text{So } p^2 + q^2 + 2\left(-\frac{b}{a}\right) = 4m^2$$

$$\therefore p^2 + q^2 = 4m^2 + \frac{2b}{a}$$

QED [2]

(iii) $N[-apq(p+q), a(2+p^2+pq+q^2)]$

$$x = -a \times \frac{b}{a}(p+q)$$

$$= b(p+q)$$

$$= b \times 2m$$

$$\therefore x = 2bm$$

$$y = a(2 + p^2 + pq + q^2)$$

$$= a\left(2 + 4m^2 + \frac{2b}{a} - \frac{b}{a}\right)$$

$$\therefore y = 2a + 4am^2 + b$$

[2]

(iv) Locus of N;

$$x = 2bm; y = 2a + 4am^2 + b$$

$$\therefore y = 2a + 4am^2 + \frac{x}{2m}$$

$$2my = 4am + 8am^3 + x$$

$$x - 2my = -4am - 8am^3$$

$$x + (-2m)y = 2a(-2m) + 4(-2m)^3$$

Compare

$$x + py = 2ap + ap^3$$

clearly this is:

(i) a straight line

(ii) a normal

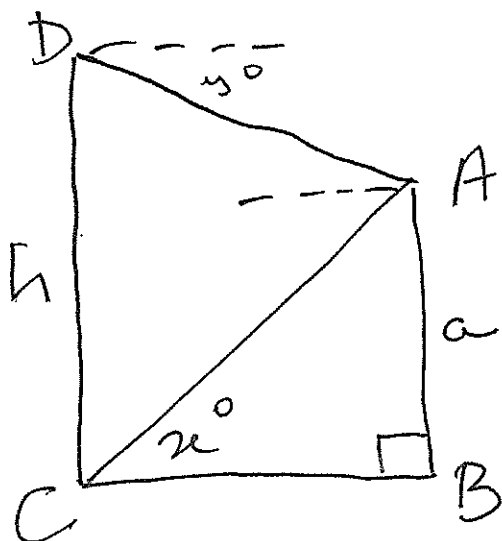
at the point where

$$t = -2m.$$

i.e. at $(-4am, 4am^2)$

[2]

(e)



$$\frac{\sin(x+y)}{h} = \frac{\sin(90-y)}{AC}$$

— Sine Rule

$$\text{But } \sin(90-y) = \cos y$$

$$\sin x = \frac{a}{AC}$$

$$\therefore \frac{\sin(x+y)}{h} = \cos y - \frac{a}{a}$$

$$\therefore h = \frac{a \sin(x+y)}{\sin x \cos y}$$

QED

[6]